Cerebral air gas embolism from concentrated hydrogen peroxide ingestion.

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INTRODUCTION: Ingestion of a small amount of concentrated hydrogen peroxide can cause cerebral air gas embolism (CAGE). Hyperbaric oxygen therapy (HBOT) is the standard of care in the treatment of CAGE. We report a case of CAGE after accidental ingestion of 33%hydrogen peroxide treated with HBOT resulting in reversal of both the clinical and radiologic abnormalities. CASE REPORT: A 48 year-old male took two sips of 33% hydrogen peroxide. A short time later, he developed hematemesis, left sided hemiplegia, confusion, and left homonymous hemianopsia. Initial laboratory studies, chest x-ray, and brain CT were normal. MRI demonstrated areas of restricted diffusion and T2 hyper intensities in multiple vascular territories consistent with ischemia due to CAGE. Eighteen hours after arrival, the patient underwent HBOT at 3 atmospheres absolute (ATA) for 30 minutes and 2.5 ATA for 60 minutes with clinical improvement. Follow-up MRI at six months demonstrated resolution of the hyper intensities. DISCUSSION: A search of MEDLINE from 1950 to present revealed only two cases of CAGE from ingestion of concentrated hydrogen peroxide treated with HBOT. Both cases, similar to ours, had complete resolution of symptoms. Of the seven reported cases of CAGE from hydrogen peroxide that did not undergo HBOT, only in one patient was there a report of symptom resolution. CONCLUSION: Ingestion of even a small amount of concentrated hydrogen peroxide can result in cerebral air gas embolism. Hyperbaric oxygen therapy may be of benefit in reversing the symptoms and preventing permanent neurological impairment.